

UDHAYA PRAKASH M *Software Engineer (Full Stack)*

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Professional Summary

Software Engineer with 2.5+ years of experience building **scalable real-time applications**, **AI-powered systems**, and **high-performance backend infrastructure**. Experienced in **React**, **Flutter**, **Node.js**, **distributed systems**, and **event-driven architectures**, with proven impact scaling products to **150K+ MAU** and achieving **sub-150ms latency**.

Experience

TerraByte Technologies *🔗*, SDE-1

July 2024 – present | Bangalore

- Scaled application from 5 to 2,000+ DAU (150K MAU) by optimizing architecture with **caching** and **asynchronous processing**, achieving under 150ms average latency.
- Reduced content production time by 75% (2 days → 0.5 day)** by building an automated, end-to-end pipeline for multi-modal content generation (text, image, video) using **LLMs** and structured prompts.
- Architected **matchmaking and real-time systems** with backend coordination, increasing multiplayer engagement by 2x.
- Implemented asset delivery system with intelligent **caching**, parallel downloads, and on-demand loading, reducing launch time from 15–20s to 4–5s and improving storage efficiency.
- Developed recommendation system using user behavioral data, improving user retention (D1: 43%, D7: 24%).
- Developed centralized **analytics system** to track user lifecycle events, creating a single source of truth and reducing codebase by 1,500+ lines.
- Built a reliable **event-driven communication** layer between Unity and React/Flutter clients, reducing synchronization failures by 60%.

Adrig Technologies *🔗*, Front End Developer (Hybrid)

January 2024 – July 2024 | Remote

- Built and deployed company website and technical documentation platform using Next.js and TailwindCSS with SSG, achieving 95+ Lighthouse score and sub-2s TTI.
- Designed **proof-of-concept frontend dashboards** for Southern Railways - **Rolling Block Management System**, implementing **RBAC** with tailored interfaces for supervisors, station masters, and back-office teams for real-time railway operations monitoring.

Legacy Code HQ *🔗*, Front End Developer

August 2023 – January 2024 | Remote

- Built frontend interface for a **code migration platform** with interactive **dependency graph visualization** and **class hierarchy mapping** for legacy **Kotlin codebases**, processing **50K+ lines of code** and accelerating migration workflows by **50%**.
- Designed and deployed responsive company website showcasing product capabilities, technical methodology; website achieved a **92 Lighthouse performance score**, boosting search engine rankings by 20%.

Technical Skills

Backend: Node.js, Express.js, TypeScript, REST APIs, WebSockets, Firebase Auth (JWT verification), Zod, HTTP/HTTPS

Frontend: React.js, Next.js, TypeScript, Redux, Flutter, Dart, Firebase (Auth & SDK), TailwindCSS, Shadcn

Databases & Caching: MongoDB, Firestore, Redis

Cloud & DevOps: AWS (EC2, S3, CloudFront, CloudWatch, Route 53), GCP (Compute Engine, Cloud Storage, CDN, Cloud Run, Cloud Build, Load Balancing, IAM), Docker

Architecture & Systems: Event-Driven Architecture, Asynchronous Processing, Caching Strategies, Real-time Systems, Performance Optimization, WebSocket Systems, Real-time Multiplayer, Microservices

AI & Data: LLM Integration, Prompt Engineering, AI-based Content Generation, Structured Output Generation, Multi-modal Pipelines (text/image/video)

Education

Vellore Institute of Technology

Bachelor of Technology - 7.88 CGPA

July 2019 – May 2023 | Chennai, India